Time, Clocks, and the Ordering of Events in a Distributed S

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Abstract & Motivation

The concept of one event happening before another in a distributed system is examined and is shown to define a partial ordering of the events. A distributed algorithm is given for synchronizing a system of logical clocks which can be used to totally order events.

The Partial Ordering Relation (->)

The relation "->" on the set of events of a system satisfies the following conditions:
1. If a and b are events in the same process, and a comes before b, then a -> b.
2. If a is the sending of a message by one process and b is its receipt by another, then a -> b.
3. If a -> b and b -> c then a -> c. Two distinct events a and b are concurrent if not a -> b and not b -> a.

Clock Condition Rules

Rule IR1: Each process P_i increments C_i between any two successive events.
Rule IR2: (a) If event a is the sending of a message m by process P_i, then message m contains timestamp T_m = C_i(a).
(b) Upon receiving message m, process P_j sets C_j greater than or equal to its present value and greater than T_m.

Applications in State Replication

Logical clocks enable totally ordered request queues across distributed nodes,
providing fair mutex and state machine replication without centralized coordinators.